# CS 7545 F26, Homework 2

100 point assignment.
120 points possible.
Points above 100 are divided by four and become bonus points.
Maximum bonus 5 points.

Due October 1, 2026

## Exercise 1 (30 points)

In class, we examined VC-dimension as an approach to achieve a generalization bound on an infinite hypothesis class $H$. Here, we'll use another technique, approximating an infinite set with a finite set, for a specific problem.

For a fixed dimension $d$, let $\mathcal{X} = [0,1]^d$ (i.e. the unit cube) and let the marginal data distribution for $x$ be uniform (i.e. the pdf has value 1 within the unit cube).

Let our hypothesis class be "boxes" with a corner at the origin: $H = \{\mathbf{21}[\forall i \in [d] : x_i \leq b_i] - 1 : b_1, \ldots, b_d \in [0,1]\}$.

We define a finite grid of hypotheses with grid resolution as an integer $R$. In particular, $B_R = \{\frac{2i-1}{2R} : i \in [R]\}$ and $H_R = \{\mathbf{21}[\forall i \in [d] : x_i \leq b_i] - 1 : b_1 \in B_R, \ldots, b_d \in B_R\}$.

Let

$$h^\star \in \arg\min_{h \in H} L(h) \tag{1}$$

$$h^\star_R \in \arg\min_{h \in H_R} L(h) \tag{2}$$

$$\hat{h}_{S,R} \in \arg\min_{h \in H_R} \hat{L}_S(h) \tag{3}$$

1. Show that

$$L(h^\star_R) \leq L(h^\star) + \frac{d}{2R} \tag{4}$$

2. Show that for any $R$, with probability at least $1-\delta$,

$$L(\hat{h}_{S,R}) - L(h^\star_R) \leq 2\sqrt{\frac{d\ln(R) + \ln(2/\delta)}{2n}} \tag{5}$$

3. Show that for an appropriate choice of $R$ (as a function of $n$ and $d$) and treating $\delta$ as constant,

$$L(\hat{h}_{S,R}) - L(h^\star) \leq \mathcal{O}\left(\sqrt{\frac{d\ln(dn)}{n}}\right) \tag{6}$$

# Exercise 2 (20 points)

Let $\mathcal{X} = \mathbb{Z}$ (the integers) and $H = \{2\mathbf{1}[a \leq x \leq a+1.5] - 1 : a \in \mathbb{R}\}$.

1. What is the VC dimension of $H$? Prove it.
2. What is $\Pi_H(n)$ as a function of $n \geq 1$? Prove it.
3. How does this compare to the upper bound of the Sauer-Shelah Lemma?

# Exercise 3 (20 points)

Suppose we have an input space $\mathcal{X}$ and two hypothesis classes $H_1$ and $H_2$ such that $d(H_1) = d_1$ and $d(H_2) = d_2$.

1. Prove that for any $n \in \mathbb{N}$,

$$\Pi_{H_1 \cup H_2}(n) \leq \Pi_{H_1}(n) + \Pi_{H_2}(n) \tag{7}$$

2. Prove that

$$d(H_1 \cup H_2) \leq d_1 + d_2 + 1. \tag{8}$$

3. Now we'll show the bound is tight. Fix $d_1$ and $d_2$. Let

$$\mathcal{X} = \mathbb{Z} \tag{9}$$
$$H_1 = \{2 \cdot \mathbf{1}[x \in A] - 1 : A \subset \mathbb{Z}, |A| \leq d_1\} \tag{10}$$
$$H_2 = \{2 \cdot \mathbf{1}[x \notin A] - 1 : A \subset \mathbb{Z}, |A| \leq d_2\} \tag{11}$$

Prove that $d(H_1) = d_1$, $d(H_2) = d_2$, and $d(H_1 \cup H_2) = d_1 + d_2 + 1$.

# Exercise 4 (20 points)

For this problem, suppose we have a sequence of increasingly "complex" hypothesis classes $\{H_k\}_{k\geq 0}$ where $|H_k| \leq \exp(k)$ and $H_k \subset H_{k+1}$. Suppose we learn on a training set $S$ by iteratively learning with an increasingly large hypothesis class until the train error $\hat{L}_S$ doesn't go down. Mathematically, let $k_S = \min\{k : \min_{h\in H_{k+1}} \hat{L}_S(h) = \min_{h\in H_k} \hat{L}_S(h)\}$ and we select $\hat{h}_{S,\text{iter}} \in \arg\min_{h\in H_{k_S}} \hat{L}_S(h)$ as the returned hypothesis. An algorithm for computing this can be seen in Algorithm 1.

**Algorithm 1** Iterative Training

**Input:** Hypothesis classes $\{H_k\}_{k\geq 0}$ with $|H_k| \leq \exp(k)$ and $H_k \subset H_{k+1}$, sampled data $S = \{(x_i, y_i)\}_{i=1}^n \sim D^n$

**Output:** hypothesis $\hat{h}_{S,iter}$

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Choose ĥ_{S,0} ∈ arg min_{h∈H_0} L̂_S(h)
for k = 1, ... do
    Compute ĥ_{S,k} ∈ arg min_{h∈H_k} L̂_S(h)
    if L̂_S(ĥ_{S,k}) = L̂_S(ĥ_{S,k−1}) then
        return ĥ_{S,iter} = ĥ_{S,k−1}
    end if
end for
```

We wish to prove the following

**Proposition 1.** *For any $\delta > 0$, with probability at least $1 - \delta$ (over the sample $S$),*

$$L(\hat{h}_{S,iter}) \leq \hat{L}_S(\hat{h}_{S,iter}) + \sqrt{\frac{k_S + \ln(n+1) + \ln(1/\delta)}{2n}} \tag{12}$$

1. Show that for any fixed $k$, with probability at least $1 - \delta$,

$$L(\hat{h}_{S,k}) \leq \hat{L}_S(\hat{h}_{S,k}) + \sqrt{\frac{k + \ln(1/\delta)}{2n}} \tag{13}$$

2. Prove that $k_S \leq n$.

3. Prove the proposition.

# Exercise 5 (30 points)

**Proposition 2.** *Fix a hypothesis class $H$ with $d(H) = d \geq 2$ and $n \geq d$. For any Algorithm $\mathcal{A}$, there exists a distribution $D$ that is realizable by $H$ where with probability at*

*least* $\frac{1}{7}$,

$$L(\mathcal{A}(S)) \geq \frac{1}{32}\frac{d}{n} \tag{14}$$

**Proposition 3.** *Fix a hypothesis class $H$ with $d(H) = d \geq 2$, $n \geq 2$, and $\exp(-n) \leq \delta \leq 1/2$. For any Algorithm $\mathcal{A}$, there exists a distribution $D$ that is realizable by $H$ where with probability at least $\delta$,*

$$L(\mathcal{A}(S)) \geq \frac{\ln(1/(2\delta))}{2n} \tag{15}$$

1. If we have $d$ points that can be shattered, we can construct a distribution where one single point gets $1-b$ probability and the remaining points get $\frac{b}{d-1}$ probability. In a sample $S$ of size $|S| = n$, in expectation, $nb$ points of $S$ fall outside the single point. Find an expression for the loss contributions from the non-single point and optimize for $b$.

2. Apply Markov's inequality to the reduction in loss on the non-single points to prove Proposition 2.

3. If we have 2 points that can be shattered, we can construct a distribution over 2 points where a single point gets $1-b$ probability and the other gets $b$ probability. Show that in a sample $S$ of size $|S| = n$, with probability $(1-b)^n$, only the single point is observed. Thus, with probability at least $(1-b)^n/2$, the other point is incorrectly classified.

4. Noting the identity $1 - x \geq \exp(-2x)$ for $0 \leq x \leq 1/2$, prove Proposition 3.